

Geometry Streaming Server Product Guide

INTRODUCTION

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Change Log

Revision	Submission Date (dd-mm-yyy)	Affected Sections or Pages	Change Summary
Initial	23-10-2019	All	Initial issue of document.

Document Overview

Identification

Property	Value
Element	IDS
Program Set	Geometry Streaming Server
Version	1.5.2

Purpose

The Geometry Streaming Server securely converts textured triangle meshes to tilesets in the 3DTiles format:

<https://github.com/AnalyticalGraphicsInc/3d-tiles>

Potentially large meshes are broken up into a hierarchy of tiles that can be scalably streamed to clients.

This software supports both generating 3DTiles tilesets and serving them to clients, and so is also referred to as both a “Mesh Tiling Server” and a “Geometry Streaming Server”.

Included in this manual are:

- instructions for deploying to Amazon Web Services
- a manual for the REST API, which is the interface for uploading input meshes and converting them to 3DTiles tilesets
- test instructions.

Overview

The Geometry Streaming Server (GSS) is an assembly under the Instrument Data System (IDS) subsystem of MGSS which provides tools and services in the area of instrument data processing. The purpose of GSS is to support the distribution and display of large 3D datasets. Specifically, GSS is a web service that accepts triangulated meshes as input and reprocesses them into a set of tiles optimized for real-time streaming. This service enables frontend display tools to interactively download and display only the portion a mesh required to fill a user’s current field of view, allowing large meshes to be displayed efficiently.

Terminology and Notation

This document uses standard terminology and notation.

Acronyms and Abbreviations:

3D	Three Dimensional
3DES	Triple Data Encryption Standard algorithm
AES	Advanced Encryption Standard
AMMOS	Advanced Multi-Mission Operations System
API	Application Programming Interface
ASCII	American Standard Code for Information Interchange
AWS	Amazon Web Services
EC2	Elastic Compute Cloud
FPS	Frames Per Second
GDAL	Geospatial Data Abstraction Library
glTF	GL (as in OpenGL: Open Graphics Library) Transmission Format
GSS	Geometry Streaming Server
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDS	Instrument Data Systems
JPG	JPEG: Joint Pictures Experts Group

JSON	Javascript Object Notation
MGSS	Multimission Ground Systems and Services
OBJ	Geometry definition file format
PLY	Polygon file format
PNG	Portable Network Graphics
REST	Representational State Transfer
S3	Simple Storage Service
SDK	Software Development Kit
SQS	Simple Queue Service
SSO	Single Sign-On
TIFF	Tagged Image File Format
TLS	Transport Layer Security
UV	“U” and “V” denote the axes of the two dimensional texture

References

Applicable JPL Rules documents:

Title	DocID
Software Development, Rev 9	57653

Applicable MGSS Documents:

Title	Document Number
AMMOS Technical Standards Profile, Rev A	DOC-001101
MGSS Implementation and Maintenance Task Requirements, Rev D	DOC-001455
Geometry Streaming Server Software Requirements Document	DOC-002000